

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Designed and deployed scalable data pipelines using **Python** and **Spark** on **AWS** to support large-scale annotation systems, processing tens of billions of events daily.
- Developed and maintained production-grade tools and services for annotated data inspection, focusing on identifying and mitigating bias and error patterns in multimedia data.
- Built interactive **data visualization dashboards** using **Power BI** to present model quality metrics and analysis results to stakeholders, enabling data-driven decision-making.
- Implemented **CI/CD** best practices and automation pipelines within a **Git** and **Jenkins** environment, reducing deployment cycles by **30%**.
- Collaborated with cross-functional teams, including data scientists, linguists, and knowledge engineers, to refine data quality and ontology services across multiple time zones.
- Managed and optimized **SQL** database schemas and queries (**PostgreSQL**) for efficient data retrieval and analysis, supporting real-time feature engineering for ML models.

SKILLS

Data Engineering – Python, Spark, SQL, Databricks, Data Pipelines, ETL, Data Warehousing, Big Data Frameworks

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Docker

MLOps & Infrastructure – CI/CD, Jenkins, Git, GitHub, API/SDK Integration, Kubernetes

Programming & Scripting – Python, Java, Spring Boot, R

Data Visualization & BI – Power BI, Tableau, DAX, Power Query

Databases – PostgreSQL, MySQL, Oracle

Development Methodologies – Agile, Scrum

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Real-Time Data Processing Pipeline for Ad Inventory Verification

- Engineered a **Python**-based data pipeline on **AWS Lambda** to process and enrich terabytes of ad inventory data daily, ensuring real-time verification of ad placements across open web and mobile platforms.
- Integrated with third-party APIs to ingest contextual data, augmenting ad content analysis and improving model accuracy by an estimated **15%**.

- Developed comprehensive monitoring and alerting using **CloudWatch** and **Splunk** to maintain pipeline health and ensure SLA compliance for critical data streams.

ML Model Performance Monitoring Dashboard

- Designed and implemented an interactive dashboard using **Power BI** and **DAX** to visualize key performance indicators and error patterns for machine learning models in production.
- Connected to a **PostgreSQL** data warehouse to aggregate model inference logs and ground truth data, providing analysts with actionable insights into model drift and bias.
- Automated data refresh processes using **SQL Server Agent** and **Python** scripts, ensuring stakeholders had access to up-to-date performance metrics.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)